GROUP 03: Effect of Image Scaling and Zoom Augmentation

WEEK 3

OBJECTIVE FOR WEEK 3: Train each of the 3 YOLO versions (YOLOv8n, YOLOv11n, YOLOv26n) separately on every augmented dataset version (Low/Medium/High scaling-zoom intensity), keeping the dataset split, image size, epochs, batch size, and hardware identical to the corresponding Week 1 baseline run.

WORK COMPLETED:

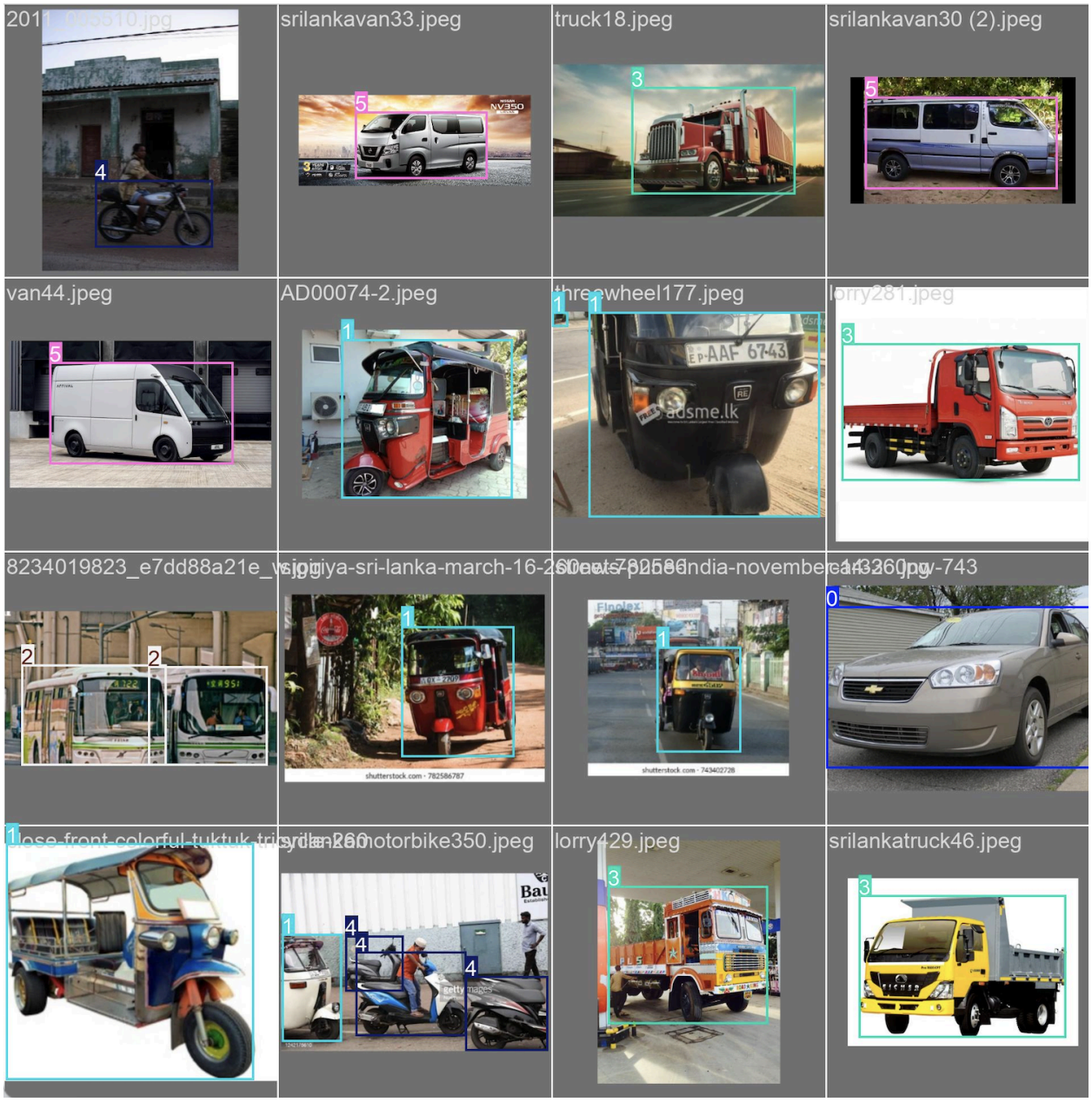
- Reloaded Week 1's baseline dataset split via Google Drive.
- Applied the Week 2 scaling/zoom augmentation to the training set, generating 3 separate datasets (Low, Medium, High); val/test kept unaugmented for fair comparison.
- Trained all 9 combinations (3 intensities × 3 YOLO versions) with the same settings as the baseline (30 epochs, 640 image size, batch 16), built-in augmentations disabled.
- Split training into separate cells per combination to avoid losing progress on session drops.
- Ran a quick validation check on all 9 models and saved their weight files for submission.

RESULTS:

1. Validation Results Table

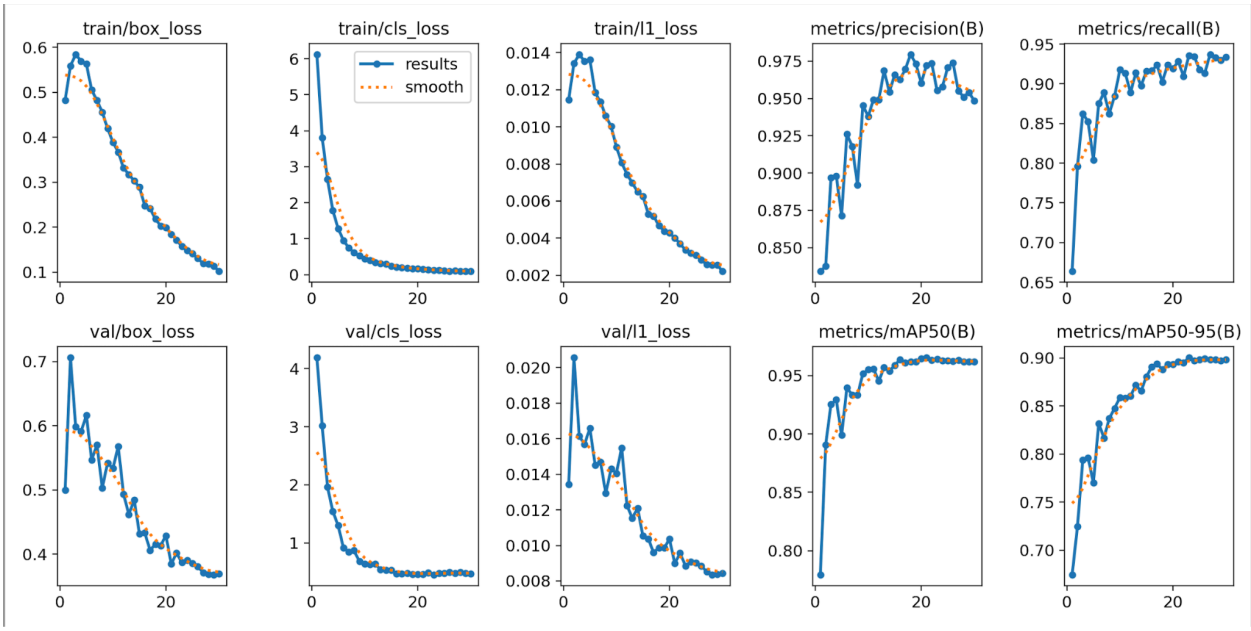
Intensity	YOLO Version	Precision	Recall	mAP50	mAP50-95	Train Time (min)	Inference Time (s, full test set)
Low	YOLOv8n	0.9455	0.8716	0.9295	0.8254	17.2	10.5
Low	YOLOv11n	0.9187	0.88	0.9217	0.814	19.1	11.4
Low	YOLOv26n	0.9717	0.9103	0.9593	0.8694	19.8	11.4
Medium	YOLOv8n	0.9398	0.8642	0.9275	0.8281	17.0	11.8
Medium	YOLOv11n	0.9388	0.8806	0.9241	0.8245	17.7	9.8
Medium	YOLOv26n	0.9596	0.8973	0.9532	0.8614	20.0	11.2
High	YOLOv8n	0.9355	0.8636	0.9206	0.8189	17.0	12.1
High	YOLOv11n	0.9555	0.8785	0.9138	0.8084	18.0	10.7
High	YOLOv26n	0.9476	0.898	0.9402	0.8606	20.3	9.6

2. Sample Detection Images

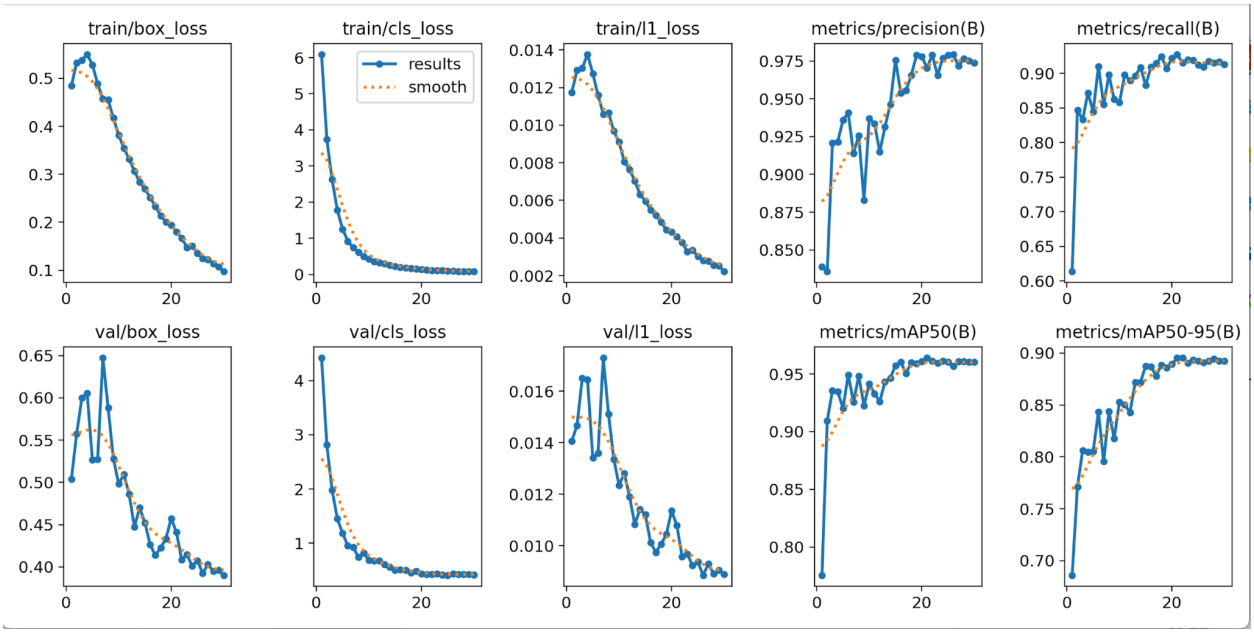


3. Training Curve Plots for a Few Runs

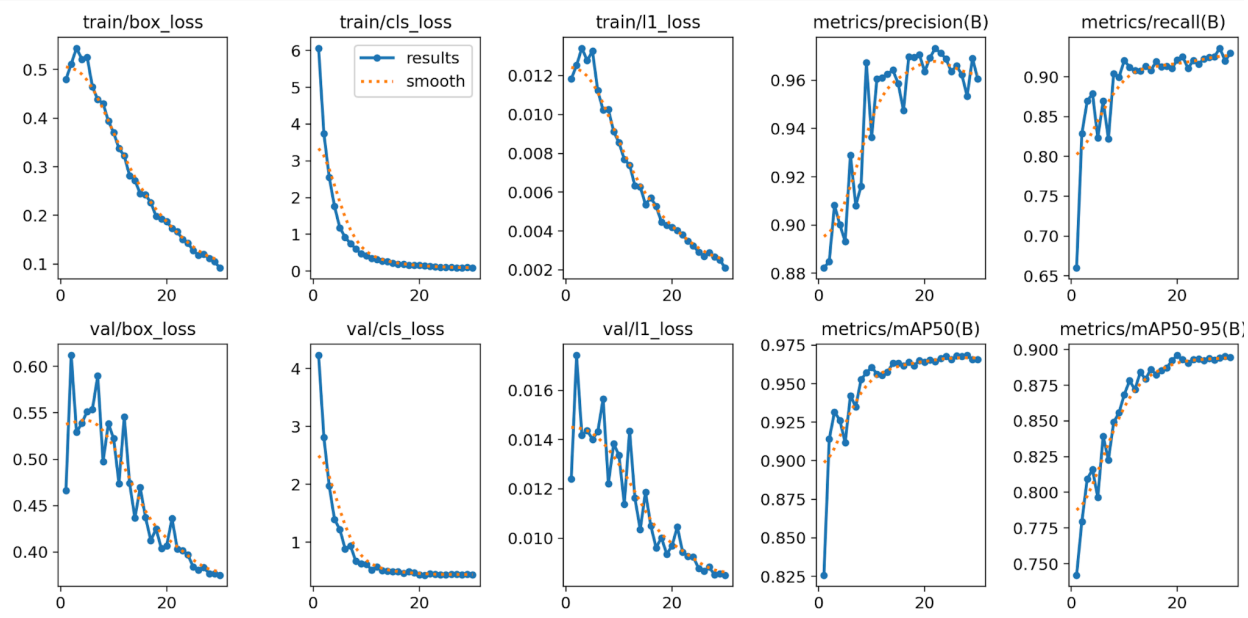
(a) High_YOLOv26n



(b) Medium_YOLOv26n



(c) Low_YOLOv26n



ISSUES/CHALLENGES:

- Hit a Drive mount error a couple of times; just retried the cell until it connected.
- Accidentally broke a path variable while editing the dataset-loading cell, which threw a NameError; had to trace back and fix it before training could start.
- The full training loop would've taken hours in one cell, so we worried about losing everything if the session dropped partway; split it into 9 separate cells so each run could be checked individually.
- Keeping track of which of the 9 combinations had finished vs. still pending took some care, especially since training each one took 15–25 minutes.

PLAN FOR NEXT WEEK:

- Consolidate results for Baseline vs. each chosen intensity level, for every YOLO version, into a single comparison matrix (intensity × YOLO version).
- Record Precision, Recall, mAP@50, mAP@50-95, training time, and inference time for every run in that matrix.
- Merge the Week 1 baseline metrics with this week's augmented-run metrics into one combined table.
- Document everything in the Week 4 report.

HOURS SPENT (PER MEMBER):

Members	Hours Spent	Contribution
Simran Koshta (B230707CS)	4	Generated the 3 augmented datasets (Low/Medium/High) by applying the scaling/zoom function to the training set, Report.
Gunna Mohana Prathyusha (B230324CS)	5	Set up and ran the training loop for 3 of the 9 model combinations, monitored for errors, Report.
Maithili Kolasani (B230385CS)	5	Set up and ran training for another 3 of the 9 combinations, Report.
Sreeneha Bhavanasi (B230092CS)	7	Set up and ran training for the remaining 3 combinations, handled the Drive mount/path issues, Report.
Ardhra Alaina Winston (B230198CS)	3	Ran validation checks on all 9 trained models, collected and zipped the weight files and results, Report.